LAUREN HENDERSON

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Research interests: ML for scientific discovery, computer vision for natural environments, rigorous model evaluation

Education

University of California, Berkeley **May 2028**
B.A. Data Science **GPA: 3.8**

- Relevant Coursework: Machine Learning, Optimization Models and Applications, Data Structures and Algorithms, Linear Algebra, Data Science, Structure and Interpretation of Computer Programs, Astronomy (ASTRON C10)
- Certifications: DeepLearning.AI Machine Learning & Deep Learning Specializations (Coursera)

Research & ML Experience

National Geographic Society (blog post | code)

Technical Project Manager (Contract) *Fall 2026 – Present*

- Promoted to lead the project's second phase after the first phase cut annotation time from months to days
- Overseeing the addition of a **supervised learning layer** to the detection and clustering pipeline, and guiding National Geographic's team in integrating the tool into a robust data pipeline for deep-sea exploration

Machine Learning Engineer (Contract) *Feb 2026 – July 2026*

- Designed and evaluated a multi-stage ML pipeline for automated marine species detection and clustering in deep-sea footage, reducing manual biodiversity annotation time from **4+ months to 1–2 days**
- Built preprocessing (median blur, CLAHE, background gating) and YOLO-World + ByteTrack detection, cutting runtime up to **86%** and achieving **100% recall** on individual animals **without labeled data**
- Fine-tuned BioCLIP-2 across **36 hyperparameter configurations** via a custom supervised contrastive loss tailored for fine-grained species differentiation, with eval metrics showing cluster integrity rise from **∼60% to 96%**
- Built the clustering stage: **HDBSCAN** over the fine-tuned embeddings (no preset species count, outliers isolated for rare species) and an **adaptive human-in-the-loop clustering** scheme that grows clusters video by video, preventing species-mixing megaclusters across **∼240 videos per mission**

UNICEF *Sep 2025 – Dec 2025*

Data Scientist (Contract)

- Built a full-stack RAG chatbot (**LangChain**, **ChromaDB**, **Streamlit**) with a **4-stage** retrieval pipeline and persona-based prompting to adapt reports for children, researchers, policymakers, and humanitarian practitioners
- Systematically identified and documented **4** recurring failure modes — persona drift, over-complex language generation, intent injection, and long-context hallucinations — and designed safety guardrails for vulnerable and young users

Research

Are Persona Vectors Language-Invariant? (write-up | code)

- Extended Anthropic's persona vectors research on **3 traits** (evil, sycophantic, hallucinating) across **English & Spanish** on Qwen2.5-3B-Instruct, finding **0.84–0.92** cross-language similarity vs. a **0.45** cross-trait ceiling
- Validated with an LLM-judged steering evaluation (**200 responses/condition**); cross-language vectors matched same-language suppression, cutting hallucinations by **up to 49%** without extracting vectors in the target language

Leadership

Data Science Society at Berkeley *Sep 2025 – Present*

Vice President, Client Relations

- Promoted from Consultant to VP overseeing DSS's largest committee (**20+ sourcers** across **3 sub-committees**); closed **5+ engagements** with Snowflake, Broadcom, National Geographic, WWF, and PG&E, procuring **$200,000+**

Skills & Interests

ML & Data: Python, PyTorch, Scikit-learn, Hugging Face, Pandas, NumPy, SQL, R, Java, Claude Code
Computer Vision: YOLO-World, ByteTrack, BioCLIP, Contrastive Learning, HDBSCAN, Video Preprocessing
LLMs & Interpretability: RAG, LangChain, ChromaDB, Fine-Tuning, Activation Steering, Prompt Engineering
Interests: Formula 1, math rock guitar, keyboard modding, outdoor bouldering, coffee, The Good Place